ClassBooker — Room Booking Management

Database project report • Diego Delgado (MySQL) and Gurleen Kaur (MS SQL Server)

Project Description

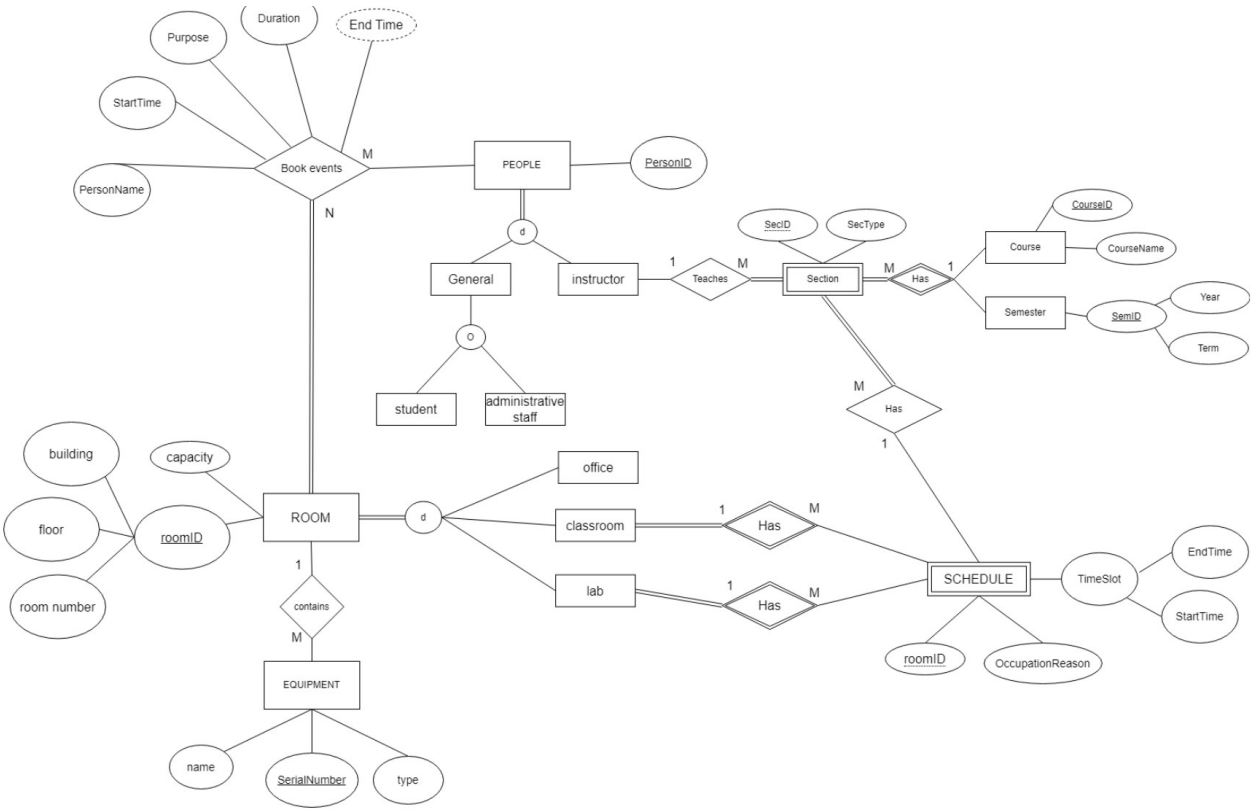
A database is created which records the information about all the classrooms on a college campus and people can reserve a room for a certain date, time and event. The data requirements for this database are summarized as follows:

- Each room is uniquely identified by a room number composed of building, floor and room number (e.g. B123 means B building, floor 1, room 23) as well as room type (office, classroom, lab…), capacity (fixed number of persons can reside in), equipment installed in. Each equipment has name, type and SerialNumber that is unique.
- If a room is a classroom or lab, it has a schedule. The schedule describes the time slot (e.g. from 10 to 12), the occupation reason (e.g. the lecture or lab of the course 2223-001), the instructor who uses the room during that time slot.
- The schedule of a room can be different for every semester.
- An instructor can teach multiple course sectors in a semester.
- People can book a room. When a room is booked, the name of the person who booked, the start time, duration, and purpose (event title) is stored in the database.

Assumptions about Cardinality and Participation

- Some classrooms have a schedule based on type. They could also be unoccupied during certain hours. Partial participation, 1:M, as all schedules must be assigned to a classroom. Total participation as all schedules must be assigned to a classroom
- Some people (like instructors) are part of a classroom schedule. Total participation from schedule to people as all schedules must have someone in attendance. Partial participation from people to schedule as not all people are required to be part of a schedule. 1:M
- If the person is an instructor, they teach courses. Partial participation as not all people are instructors. Total participation from courses, as all courses must be taught by an instructor. 1:M
- People create bookings. Partial participation as not all people create bookings. Total for bookings as all bookings must be created by people. 1:M
- Bookings reserve classrooms. Total participation for bookings as all bookings reserve a classroom. Partial for classrooms as not all classrooms are reserved by bookings. M:1
- One classroom can have multiple equipment, but one equipment can only be placed in one classroom. Partial participation, 1:M

EER Diagram



Relational Schema (ER-to-relational mapping, after normalization)

Everything has been normalized as we cannot further break down the tables. We have checked 1NF, 2NF, 3NF. Nothing is referenced indirectly and no attribute is being referenced by partial keys.

Course (CourseID, CourseName)
Semester (Year, Term)
Schedule (ScheduleID, OccupationReason, (Building, Floor, RoomNumber))
TimeSlot (ScheduleID, StartTime, EndTime, DayOfWeek)
Equipment (SerialNumber, Name, (Building, Floor, RoomNumber))
EquipmentName (Name, Type)
People (PersonID, FirstName, LastName)
Instructor (PersonID)
General (PersonID)
Student (PersonID)
Administrative_Staff (PersonID)
Course_Semester_Section ((Year, Term), CourseID, SecID, SecType, ScheduleID, PersonID)
People_BookEvents_Room (PersonID, (Building, Floor, RoomNumber), StartTime, Purpose, Duration, Date)
Room (Building, Floor, RoomNumber, Capacity)
Office (Building, Floor, RoomNumber)
Classroom (Building, Floor, RoomNumber)
Lab (Building, Floor, RoomNumber)

Data Types and Constraints

We mostly used int and Varchars. Most of the data within our database is number based like all the IDs, while names are just secondary information. In trying to duplicate Langara's building system we limited the Building to CHAR(1) as it would be A, B, C, etc. For the rest of the numbers most of them would vary in size so we had to select int for the vast majority of them.

In TimeSlot for DayOfWeek we chose int, as we are planning on mapping 0-6 as Mon-Sun and then associate specific slots to schedules that repeat on certain days. That way we can simplify the table and only have the existing slots rather than a row for every day of the week.

Scripts

The SQL scripts for creating the tables, inserting the sample data, and the 20 queries and view are in the scripts folder: Create.sql, Insert.sql and Query.sql.